

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

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paper doi:10.5281/zenodo.20706893 · code github.com/Vortx-AI/emem (Apache-2.0) · live emem.dev · method: whitepaper v3, emem @18adb67

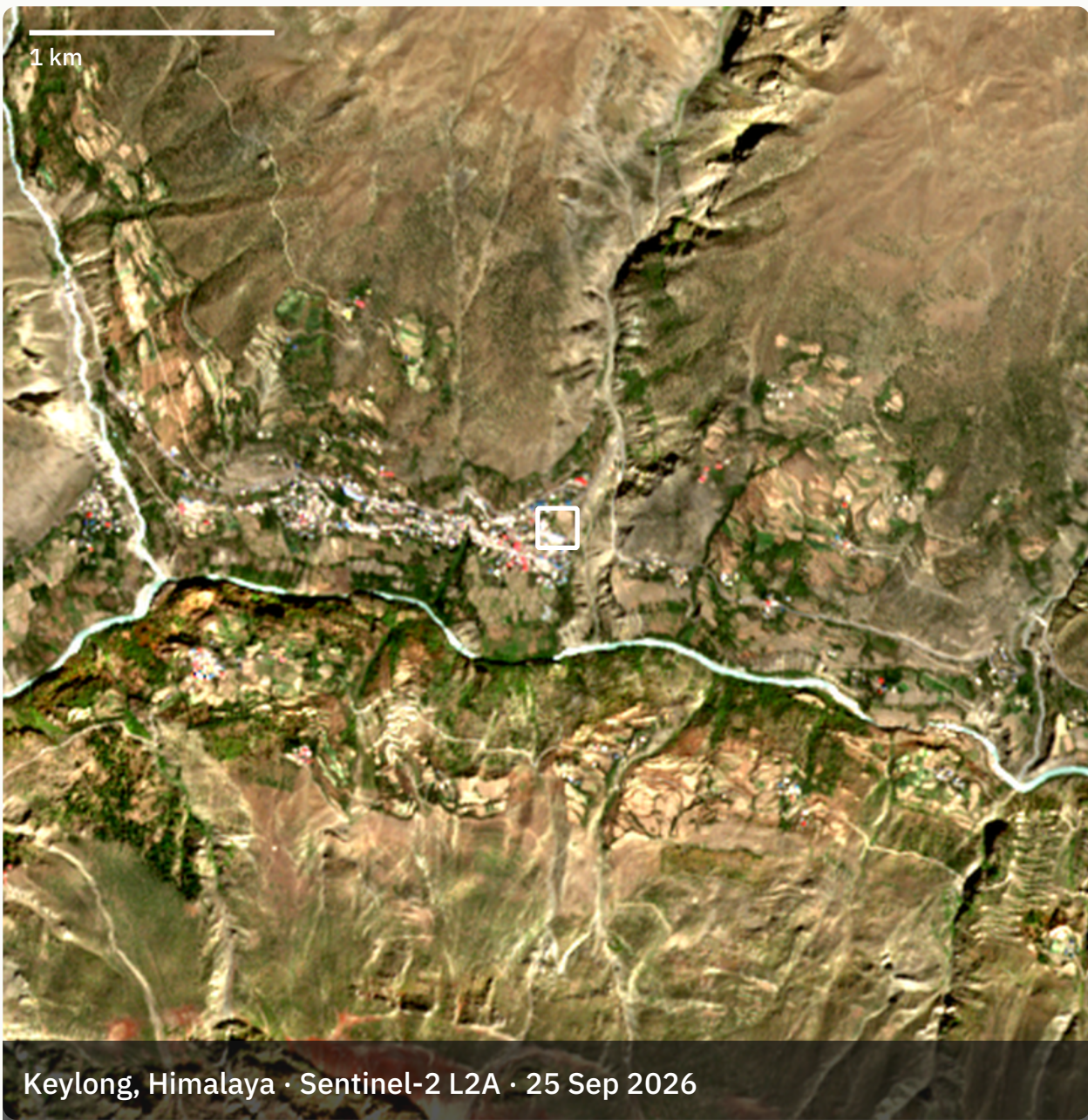


THE PROBLEM

Agents that work together drift apart.

Each agent reads the world through its own searches and summaries. After a few handoffs two agents no longer agree on where a place is, what was measured or when, and neither can show the other which number is right. Agreement does not rescue them (R2).

THE WORLD WRITES



a 2.02 GB scene; the square marks the cell of the NDVI record (drawn 100 m wide)

OTHER WRITERS

Copernicus DEM · Overture Maps · weather · TESSERA (retired) · device gate: ships, no hardware enrolled yet

114

MCP tools; 18 in the core loop; no key

168

published algorithms: flood, heat, crop, solar

118

wired measurements, 46 source schemes

2.57 M

signed log entries, 30 Sep 2026

177

REST paths in the live OpenAPI

10

client paths, one fact_cid, three runs each

Six contributions

each a mechanism in emem's code at 18adb67, each with a measurement on the live system

C1 An address for every observation

A value is keyed by where, what and when, and named by the hash of its exact bytes. The key finds a record; the name proves which one.

```
k = (cell164, band, tslot)
fact_cid = base32(BLAKE3-256(CBOR(f)))
token = emem:fact:<cell>:<fact_cid>
```

A **1,115-byte** record read from a **2.02 GB** scene travels as **84** characters **§4**; ten client paths re-derive the same fact_cid.

C4 One token algebra, pixel to archive

The same naming covers a value, a field, a field through time, a set of fields, any file hashed where it lies, and one handle for many facts. Each family declares how strongly it binds.

```
fact → raster → cube → rasterset
tree = Merkle(chunks of a file, in place)
bundle = one handle over ≤ 256 fact_cids
```

A 443 × 453 px B04 field under one token **§5**; a 5-scene cube whose members name the scene actually used **§6**; a **238.7 MB** COG hashed in 292 chunks without moving it **§2**; 8 records behind one 38-character line **§16**.

C2 A memory on two clocks

Valid time (when the Earth was seen) and record time (when it was signed) are both in the hashed bytes. New evidence appends; nothing is rewritten.

```
as of t: argmax_signed_at ≤ t (signed_at, cid)
receipt binds (as_of_tslot, as_of_signed_at)
log leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation)))
```

Bengaluru elevation moved **918.0 → 915.07 m** when the upstream DEM changed **§12 §13**. The May token still verifies, "as of 15 Jun" still returns **918.0**, and the pair is flagged as a provider change.

C5 Verification that needs no trust

A receiver checks bytes, place, arithmetic, source pixel, signature and log with stock libraries. Each record shows how deep it can be checked.

```
accept ⇔ BLAKE3(b) = cid ∧ cell(b) = cell(token) ∧ NDVI(DN(b)) = value ∧ COG(row, col) = DN(b) ∧ Ed25519(o) ∧ leaf ∈ STH
depth: recomputable > rerunnable > trace-verified > checkpoint-signed > attester-only
```

Our script: **15 links**, **17 checks**, **17.9 s**, no emem code **§22**. An unaffiliated agent's clean-room verifier from /v1/verifier_spec: **725** requests, zero errors, five tampered receipts rejected, eight defects found and fixed.

C3 Machines write facts; agents write claims

Only emem's readers of registered archives, enrolled devices and operator-listed keys write the fact plane. Agents sign notes under their own keys: data, never facts, never instructions to the reader. When nothing qualifies, emem signs an absence.

```
cell(bytes) ≠ cell(token) → refused (HTTP 409)
absence = (cell, band, tslot, reason_cid, sources)
```

Venice: *no carriageway within 50 m* in Overture 2026-09-23.1, with the row groups it read (exhg61ps...). /v1/p/ane/conformance tests that no tool takes a caller's value.

C6 Computation others can recompute

An agent derives a value over signed facts and emem re-runs the pure operation before recording it. A guard refuses a sentence whose number disagrees with the fact it cites.

```
derive accepted ⇔ ulp_gap(v_caller, v_emem) = 0 (≤ 4 for n-ary sum, mean)
guard: stated ≠ signed at the written precision → deny
```

Our NDVI change of **−0.1298** (25 Sep 2025 → 2026) was recomputed bit-identical and signed under our key **§11**. The guard refuses "35 °C" citing a signed 28.0 °C **§14**.

What agents do with it

eight jobs served today by the same tools; one memory for every model

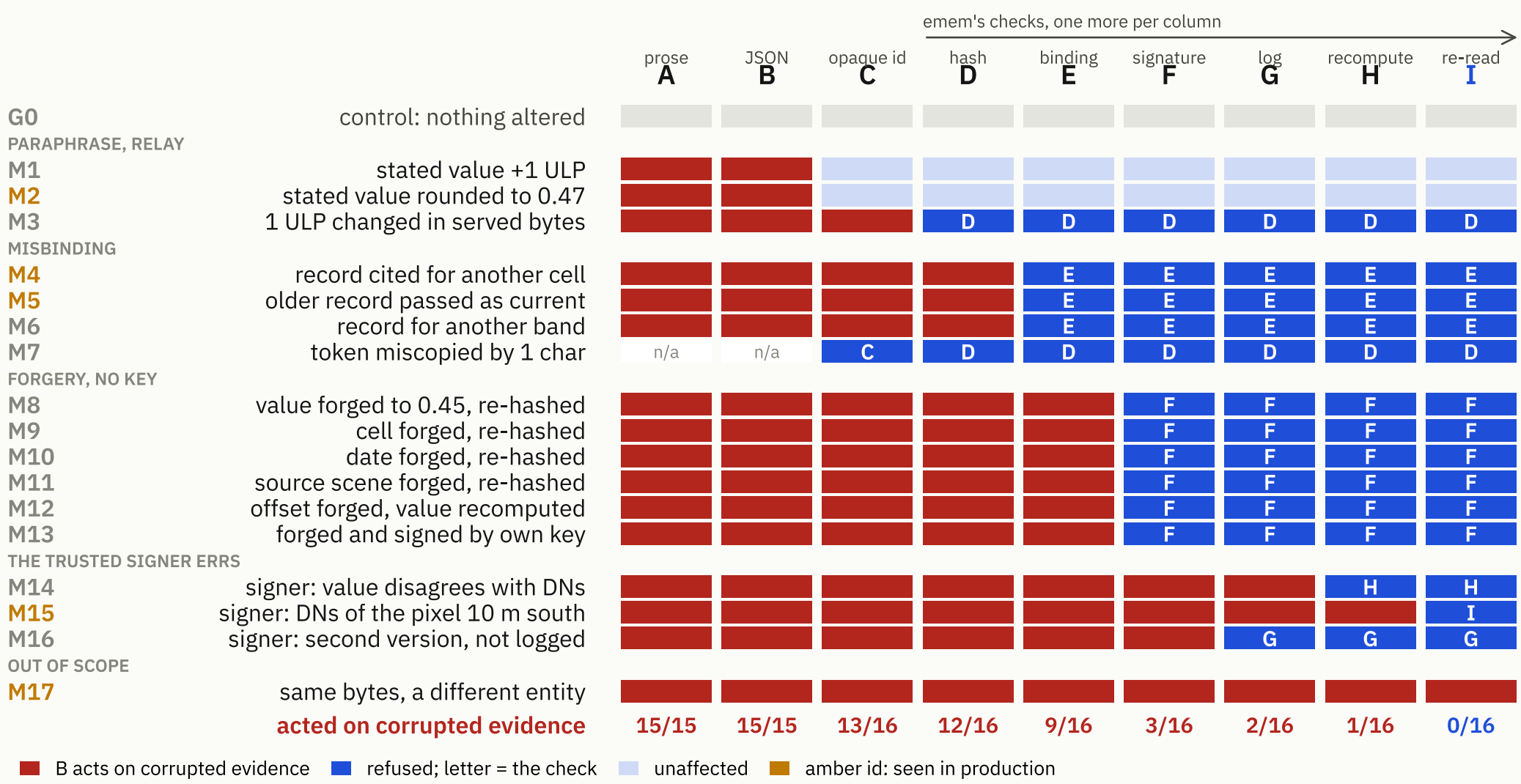
Research a place, no web <code>ask · 168 recipes</code> Flood, heat, crop or solar scores from signed readings; each recipe cited by id.	Hand work to another agent <code>emem:bundle</code> Exact signed bytes behind one line, on any model or vendor.	Keep long work alive <code>emem_memory_create</code> Signed notes outlast sessions, compaction and restarts. Notes are public.	Agree on what a thing is <code>emem_entity_resolve</code> A farm or a building gets one identity; different phrasings converge on it.	Explain why a number moved <code>emem_change_attribution</code> Names the terms behind a change, each with its fact ids.	Catch a contradiction <code>emem_memory_contradictions</code> Finds records that disagree; the guard refuses a wrong number.	Turn documents into evidence <code>emem:tree</code> Lab reports and land records become signed units under one token.	Compute for recomputation <code>emem_derive</code> Pure operations are re-run by emem before the result is recorded.
One token, three vendors Claude (through MCP), Gemma 3 on Amazon Bedrock and Qwen 2.5 running locally all end with the same fact_cid and value. The decoding is emem's, not the model's: two of the three were handed the same resolved response.		A public dispute, settled by bytes geo.qa's agent re-derived a Doha road fact and reported 9.8 m against emem's 5.4 m. Re-measuring from the full-precision coordinate inside the fact gave 5.4 m, and that agent withdrew its number in a signed note.		Regulation an auditor can re-check eudr.dev checks farm plots against the EU Deforestation Regulation cut-off with emem's forest facts, and prepares Annex II statements that an auditor can re-verify.		Any node, the same receipts The hosted node runs the Apache-2.0 binary, and a receipt minted on one node verifies on another. geo.qa runs a second node; an air-gapped build runs with no network at all.	

Four results

scripts, data and pre-registration hashes: github.com/Vortx-AI/esa_poster/research/repro

R1 · THE MUTATION TEST: 17 CASES × 9 DEPTHS, ON REAL BYTES

Each check stops a failure that the others let through

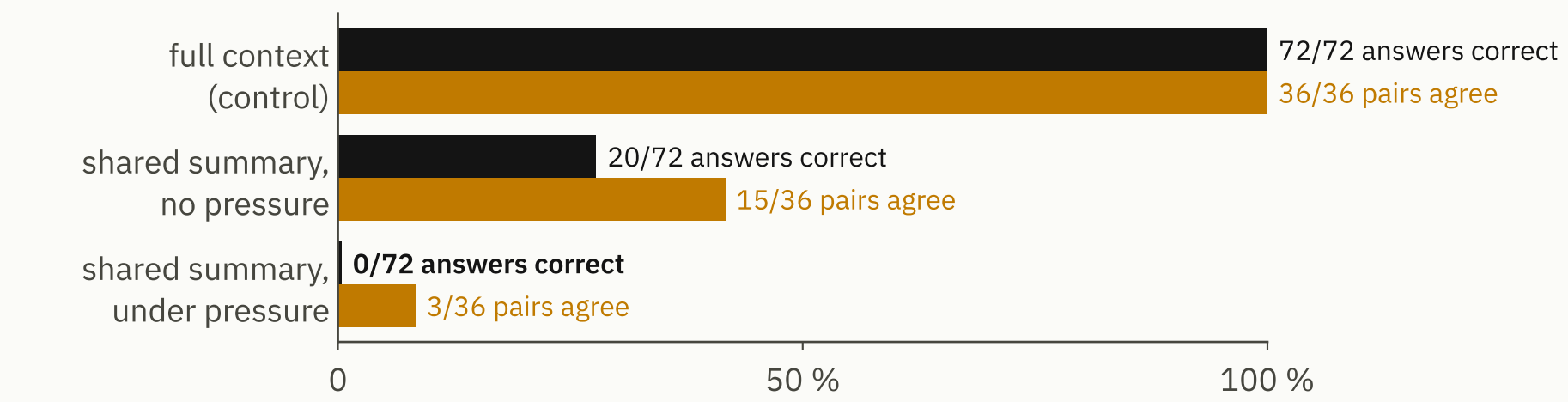


B receives the Keylong record **§4** after one corruption; a deterministic verifier, no model in the loop, decides whether B acts on it. The rule "irrigate iff NDVI ≤ 0.4705" flips in 6 of 16 cases. T2 rows use a test key the verifier trusts. Leave-one-out: dropping binding, signature, log, recompute or re-read each lets a case through; the signature subsumes the hash. Units, checkpoints and embeddings are not covered.

Prose and JSON pass all 15 applicable corruptions. Full emem passes none, and still cannot tell which entity A meant.

R2 · COMPACTION AND HANDOFF

Agreement is not evidence



Pre-registered; Gemma-4-12B and Qwen2.5-7B on one host **§17**. Under pressure, 3 of 36 pairs agreed while 0 of 72 answers were right (Fisher p = 0.035; pairs share trials, so descriptive). In a separate handoff study, n = 20 per arm **§18**, the exact value survived prose 2/20, dense retrieval 8/20, BM25 20/20, the emem bundle 20/20.

Agreement between agents cannot certify a value.

R3 · TWO LLMS, EMEM THE ONLY TOOL

A token helps only when checked

B RECEIVES	B RIGHT	B CHECKED
A's token	10/10	resolved; harness re-hash, receipt 10/10
A's prose, 16 digits	10/10	nothing to check
prose rounded to "0.47"	0/5	nothing to check
token with a forged cell	5/5 declined	MCP error; REST 409

A = Claude Sonnet 5.5, B = Claude Haiku 4.5, pre-registered **§22**; token vs full prose tied (p = 1); rounded arm exploratory. A Qwen2.5-3B arm did worse; its logs are not yet committed.

The check belongs in the resolver, not in the model's goodwill.

R4 · FALSIFIABLE BY CONSTRUCTION

The records located emem's own error



Re-reading the scenes named inside the records showed that before 28 Sep, **162 of 200** sampled Sentinel-2 records (81 %, Wilson 95 % CI 75–86 %) carried the DNs of a neighbouring pixel; here, 10 m south **§7**. emem rounded a fractional pixel index where GDAL takes the floor. After the fix **§8**, 49 of 49 records where the two rules differ read the containing pixel. No signed byte was rewritten, so every affected record stays resolvable and datable.

Errors stay findable, datable and bounded.

Objections we expect

and our answers

A signature does not make a value true.

AGREED It proves who served which bytes, and when. Truth is checked by re-reading the public source each record names. That is how R4 was found.

A token costs more LLM tokens than the number it names.

AGREED About 6× for one value in our count (46 vs 8 tokenizer tokens). A single token buys identity, not compression. The bundle compresses: up to 256 facts behind one 38-character line.

Is this a guardrail, provenance-first composition, or STAC with checksums?

REFRAME Different layers. A guardrail such as GeoGuard decides whether a claim should stand; provenance-first composition traces a pipeline; STAC, C2PA and PROV locate files, sign media or describe lineage. emem names the exact observation one agent cites, so another can verify it. A guardrail can take emem tokens as its evidence.

One operator runs the responder and the log.

AGREED 111 keys co-sign the log, but the only domain-vouched one, geo.qa, is also Vortx AI. Every check above runs without us, and an unaffiliated clean-room verifier reproduced them (C5). Independent operators are the next step.

The title says foundation-model embeddings, yet the encoders were retired.

REFRAME Clay, Prithvi-EO-2.0, Galileo and a JEPHA head left the code after acceptance **§23**, and emem.dev now mints no new TESSERA vectors. Vectors signed before still resolve and verify **§24**: the memory outlived its encoders.

BM25 retrieval matched you.

REBUTTAL On exactness, when the string is in the corpus (16/16). Retrieval returns text nobody can check; dense retrieval gave 4/142 exact and up to 138 confidently wrong, by a median 252 m **§25**.

Your own reader signed wrong-pixel records.

AGREED 162 of 200 before 28 Sep. The records made the error locatable by re-reading the scene they name, and datable by their signed_at. A store of bare vectors could show only that a value moved.

Small n, one host.

AGREED The agent studies are samples: pre-registered, hashed before trial 1, not yet replicated across models. The mutation test (R1) is deterministic and runs in a second from the repository.

THE INVENTION

One place, one address. One observation, one signed record. The record's name, not a paraphrase, is what crosses between agents.

A memory of the physical world that agents share and no agent controls: **written by machines, named by the hash of its bytes, kept on two clocks**. In a 17-case mutation test on real bytes, prose passed all 15 corruptions it can carry; the full check chain passed none (R1).